10. C(diamond) → C(graphite), ΔH = –ve, this indicates that

a) graphite is more stable than diamond

b) graphite has more energy than diamond

c) both are equally stable

d) stability cannot be predicted

11. The enthalpies of formation of Al_2O_3 and Cr_2O_3 are – 1596 kJ and – 1134 kJ, respectively.

ΔH for the reaction $2Al + Cr_2O_3 \rightarrow 2Cr + Al_2O_3$ is

a) – 1365 kJ b) 2730 kJ c) – 2730 kJ d) – 462 kJ

12. Which of the following is not a thermodynamic function ?

a) internal energy b) enthalpy c) entropy d) frictional energy

13. If one mole of ammonia and one mole of hydrogen chloride are mixed in a closed container to form ammonium chloride gas, then

a) $\Delta H > \Delta U$ b) $\Delta H - \Delta U = 0$ c) $\Delta H + \Delta U = 0$ d) $\Delta H < \Delta U$

14. Change in internal energy, when 4 kJ of work is done on the system and 1 kJ of heat is given out by the system is

a) +1 kJ b) – 5 kJ c) +3 kJ d) – 3 kJ

15. The work done by the liberated gas when 55.85 g of iron (molar mass 55.85 g mol^{-1}) reacts with hydrochloric acid in an open beaker at 25^0 C

a) – 2.48 kJ b) – 2.22 kJ c) + 2.22 kJ d) + 2.48 kJ

16. The value of ΔH for cooling 2 moles of an ideal monatomic gas from 125^0 C to 25^0 C at constant pressure will be $\left[\text{given } C_P = \frac{5}{2} R\right]$

a) – 250 R b) – 500 R c) 500 R d) + 250 R

17. Given that $C(g) + O_2(g) \rightarrow CO_2(g)$ $\Delta H^0 = -$ a kJ; $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$ $\Delta H^0 = -$b kJ; Calculate the ΔH^0 for the reaction $C(g) + \frac{1}{2}O_2(g) \rightarrow CO(g)$

a) $\frac{b+2a}{2}$ b) $2a-b$ c) $\frac{2a-b}{2}$ d) $\frac{b-2a}{2}$

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